MIL-L-13738 (Ord)
17 November 1954
SUPERSEDING
PA-PD-388
15 June 1953

MILITARY SPECIFICATION

LEAD SALICYLATE

1. SCOPE

1.1 This specification covers lead salicylate for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366	Sieves, Standard, For Testing Purposes

MILITARY

MIL-G-2550	General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publication.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78	- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950), available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)".)

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3. REQUIREMENTS

3.1 Form.-Soft crystals.
3.2 Color.- White.
3.3 Lead content.-43.0 percent, minimum.
3.4 Salicylate content.-55.4 percent, minimum.
3.5 Volatile matter.0.25 percent, maximum.
3.6 Granulation.-Not more than 0.1 percent of the lead salicylate shall be retained on a U. S. Standard No. 325 sieve.
3.7 Workmanship.-The lead salicylate shall be free of foreign material such as dirt, grit and other extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material from the same batch or blending operation, subjected to the same processing operations and conditions.
4.2 Sampling.-Ten percent, but in no case more than 10 nor fewer than 3, of the containers, shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer, plant, contract or purchase order number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the specification. All primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements.
4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.
4.4 Determination of lead content.-Transfer an accurately weighed portion of approximately 1 gram of the sample to a 150-ml beaker. Add 10 ml of distilled water, and then add slowly, with stirring, 20 ml of a solution consisting of equal parts by volume of concentrated nitric acid and concentrated sulfuric acid. Place the beaker and contents on a hot plate and heat until dense white fumes of sulfur trioxide are evolved. Allow the beaker to cool for about 2 minutes, add 5 ml of concentrated nitric acid, and heat again until dense fumes of sulfur trioxide are evolved. Repeat the addition of nitric acid and heating to fumes until all the organic matter is destroyed and the sulfuric acid solution is practically colorless. Remove the beaker from the hot plate and allow it to cool. Wash down the side walls of the beaker with about 10 ml of distilled water, and again heat to fumes of sulfur trioxide. Allow the beaker and contents to cool, then cautiously dilute to about 100 ml with distilled water. Again allow the mixture to cool, add 25 ml of ethyl alcohol, and let stand for at least 2 hours. Filter the precipitate of lead sulfate, using a tared gooch type crucible. Transfer the precipitate to the crucible with a stream of 1:20 sulfuric acid solution directed from a wash bottle. Wash the precipitate three times with the sulfuric acid solution and twice with 95-percent ethyl alcohol. Aspirate the crucible

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and contents until the odor of alcohol is no longer detectable with the vacuum line turned off. Dry in an oven for 30 minutes at 100° to 105°C, ignite in a muffle furnace at 550° to 600°C for 15 minutes, cool in a desiccator and weigh. Calculate the gain in weight of the crucible to percent lead in the sample as follows:

$$\text{Percent lead} = \frac{68.32A}{W}$$

where:

A = gm of ignited residue
W = gm of sample

4.5 Determination of salicylate content.-Transfer an accurately weighed portion of approximately 0.20 gram of the sample to a 500 ml iodine flask. Dissolve the sample in 10 ml of 10% sodium hydroxide solution, and dilute to approximately 150 ml with distilled water. Add 50 ml of 0.2 N bromide-bromate solution (5.6 gm $KBrO_3$ and 15 gm KBr per liter), measured accurately from a pipette, to the contents of the flask. Add 25 ml of 1:1 hydrochloric acid solution, stopper the flask, and pour a few ml of 10% potassium iodide solution around the stopper. Allow to stand for 30 minutes, carefully loosen the stopper, and add 25 ml of 10% potassium iodide solution to the contents of the flask by pouring it around the loosened stopper. Titrate the liberated iodine with standard 0.2N sodium thiosulfate solution, adding starch as an indicator near the end of the titration. When close to the end point, stopper the flask and shake vigorously after the addition of each drop of titrant. At the end point, the suspension should be bright yellow in color. Conduct a blank determination using 50 ml of the 0.2N bromide-bromate solution exactly as described above except that the sample will be omitted. Calculate the salicylate content as follows:

$$\text{Percent salicylate} = \frac{(A-B) \times N \times 2.285}{W}$$

where:

A = ml of sodium thiosulfate solution used for titration of blank
B = ml of sodium thiosulfate solution used for titration of sample
N = normality of sodium thiosulfate solution
W = grams of sample

4.6 Determination of volatile matter.-Transfer an accurately weighed portion of approximately 5 grams of the sample to a tared glass stoppered weighing bottle approximately 60 mm x 30 mm. Dry in an oven with the stopper off for 2 hours at 100° to 105°C, replace the stopper, cool in a desiccator, and weigh. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{A \times 100}{B}$$

where:

A = loss in weight of sample
B = grams of sample

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4.7 Determination of granulation.-Transfer 50 grams of the sample to a 600 ml beaker and add approximately 300 ml of 1% aerosol solution previously saturated with lead salicylate. Stir until the sample is thoroughly wetted, then pour the suspension on a No. 325 U. S. Standard sieve complying with Specification RR-S-366. Transfer any material adhering to the walls of the beaker to the sieve, using a stream of the aerosol solution directed from a wash bottle. With the aid of a soft brush and a stream of the aerosol solution agitate the sample until no more material passes through the sieve. Wash any material retained on the sieve with distilled water, saturated with lead salicylate, and dry the sieve and contents in an oven for two hours at 100° to 105°C. Transfer the residue to a tared weighing dish, cool, and weigh. Calculate the percent retained on the sieve.

4.8 Resubmission and retest.-If the composite sample or any primary sample subjected to test, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance.. The resubmitted lot shall be accepted, provided that the new samples, selected in accordance with 4.2, pass all the tests required by this specification,and that 3 additional primary samples pass any test or tests failed by the material on original submission.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Unless otherwise specified lead salicylate shall be delivered in standard commercial containers, so constructed and protected as to conform with Interstate Commerce Commission Regulations for Transportation of Explosive and Other Dangerous Articles, etc., as applicable, at the lowest rate, to the point of delivery.

5.2 Marking.-In addition to any special marking required by the contract or purchase order the shipments shall be marked in accordance with Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Articles, etc. and in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

NOTICE.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0-8279

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